



NATIONAL OPEN UNIVERSITY OF NIGERIA
PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWAY, JABI-ABUJA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
2025_2 EXAMINATIONS

COURSE CODE: CIT332
COURSE CREDIT: 3 Units
COURSE TITLE: Survey of Programming Languages
TIME ALLOWED: 2½ Hours

INSTRUCTION: Answer Question One (1) and any other Three (3) Questions

Question 1

- 1a. Describe six (6) features of High-Level languages. **(6 Marks)**
1b. Differentiate between Low Level and High Level Languages by filling in the blank spaces in the table below. **(6 Marks)**

	Parameter	High-Level Language	Low-Level Language
1.	Basic		
2.	Ease of Execution		
3.	Process of Translation		
4.	Efficiency of Memory		
5.	Portability		
6.	Comprehensibility		

- 1c. List four (4) key concepts of imperative programming languages. **(2 Marks)**
1d. Define Logic/Declarative/non-algorithmic based languages. **(4 Marks)**
1e. Identify any four (4) key concepts of functional programming. **(4 Marks)**
1f. Describe the key concept of Scripting Languages called Dynamic Typing. **(3 Marks)**

Question 2

- 2a. Define Scripting Languages. **(1 Mark)**
2b. Highlight any two (2) roles of programming languages. **(3 Marks)**
2c. Expressivity is an evaluation criteria that means the ability of a language to clearly reflect the meaning intended by the algorithm designer (the programmer). Identify and describe three (3) other Language evaluation criteria. **(6 Marks)**
2d. Discuss the importance of Evaluation in programming. **(3 Marks)**
2e. Identify two (2) key core concepts of Logic programming. **(2 Marks)**

Question 3

- 3a. Highlight four (4) features of Low Level Languages. **(6 Marks)**
3b. Define Parsing. **(2 Marks)**
3bii. Draw a well-labeled flow diagram of the Parsing process. **(7 Marks)**

Question 4

- 4a. Describe two (2) approaches to Parsing. **(4 Marks)**
4b. Outline five (5) key concepts of Functional Programming. **(5 Marks)**
4c. Explain four (4) Categories of Expressions used in programming languages. **(6 Marks)**

Question 5

- 5a. Define Syntactic Ambiguity. **(2 marks)**
5b. Selection Logic involves a number of conditions or parameters which decides one out of several written modules. The structures which use this type of logic are known as Conditional Structures. Provide sample code for the following two (2) structures: **(6 Marks)**
 i. Double Alternative **(2.5 Marks)**
 ii. Multiple Alternatives **(3.5 Marks)**
5c. Describe the following concepts in programming languages: **(7 Marks)**
 i. Static Scope
 ii. Dynamic Scope
 iii. Referencing

Question 6

- 6a. Define Type Checking. **(2 Marks)**
6aii. Differentiate between Strongly Typed and Weakly Typed Programming Languages. **(6 Marks)**
6b. In the context of Abstracting Concepts using Procedures, outline four (4) principle actions associated with a procedure call/return mechanism during implementation. **(4 Marks)**
6c. Outline three (3) techniques used to interpret the syntax of a programming language. **(3 Marks)**



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2024_2 EXAMINATION.**

COURSE CODE: CIT332

COURSE CREDIT: 3

COURSE TITLE: SURVEY OF PROGRAMMING LANGUAGES

TIME ALLOWED: 2½ HOURS

INSTRUCTION: ANSWER QUESTION ONE (1) AND ANY OTHER THREE (3) QUESTIONS

QUESTION

- 1(a) In a tabular form, differentiate between C++ and Python languages [6 marks]
- (b) Distinguish between a language recognizer and a language generator [4 marks]
- (c) In designing hotel accommodation reservation; explain why you would trade-off adaptability to Reliability [7 marks]
- (d) Establish the link between a class, object, attributes and methods of an entity named STUDENT in a result computation system [8 marks]

- 2 (a) Highlight the four (4) phases of interpretation [4 marks]
- (b) Evaluate the risks involved if the criteria of using a programming language are violated [7 marks]
- (c) Identify four (4) features of an algorithm [4 marks]

- 3 (a) Appraise the strength of Object Oriented programming languages [6 marks]
- (b) Differentiate between recursion and iteration [4 marks]
- (c) Identify 5 important features used for manipulating expression in LISP programming [5 marks]

- 4 (a) Consider $G = (\{S, A, B\}, \{a, b\}, S, P)$ $N = \{S, A, B\}$
 1. $P: S \rightarrow AB$
 2. $A \rightarrow Aa$
 3. $B \rightarrow Bb$
 4. $A \rightarrow a$
 5. $B \rightarrow b$Derive the string aabbbb from the grammar [10 marks]
- (b) Show diagrammatically the processes of hybridizing interpretation and compilation [5 marks]

- 5 (a) Differentiate between dynamic semantic and static semantic [4 marks]
- (b) Write a program segment to print the score of 10 students using the FOR-TO loop in C ++ [4 marks]
- (c) Examine the functions of the following Java operators and give two examples of each (i) Arithmetic (ii) Relational (iii) Bit-wise (iv) Logical [7 marks]

- 6(a) Differentiate between Top-down parser and Bottom-up parser [6 marks]
- (b) Identify 4 naming conventions for LISP programming language [4 marks]
- (c) List 5 features of ALGOL 68 programming language [5 marks]



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2024 1 EXAMINATION**

COURSE CODE: CIT332

COURSE CREDIT: 3

COURSE TITLE: SURVEY OF PROGRAMMING LANGUAGES

TIME ALLOWED: 2½ HOURS

INSTRUCTION: ANSWER QUESTION ONE (1) AND ANY OTHER THREE (3) QUESTIONS

QUESTIONS

- 1(a) Tabulate three (3) main similarity features of Python and Java programming language that make them a programming language of choice in the fifth generation. [6 marks]
- (b) How is Von Neumann architecture related to programming [6 marks]
- (c) Appraise the role of user friendliness in the development of computer aided learning for kindergartens [4 marks]
- (d) Analyse PEARL single-line and multiple- line comments and illustrate each with an example [4 marks]
- (e) What is meant by a programming language being machine independent [5 marks]
- 2 (a) Write a simple C++ program to print the statement “Nigeria shall rise again” followed with an in-line comment [3 marks]
- (b) Give the syntax for each of the following in C sharp (C#) (i) While loop (ii) For-loop (iii) Go to loop [9 marks]
- (c) List 3 reasons for studying programming languages [3 marks]
- 3 (a) Draw a parse tree for the expression $G = B*(C+A)$ [8 marks]
- ((b) Write a C++ program using the switch statement to determine grades using the letters A, B, C or F depending on whether variable k has the value 1,2,3 or 4. Otherwise a warning of out of range is flagged [7 marks]
- 4 (a) Consider $G = (\{S, B < C\}, \{a, b, c\}, p, s \}$ where the productions p are)
1. $S \rightarrow aSBC$
 2. $S \rightarrow abC$
 3. $CB \rightarrow BC$
 4. $bB \rightarrow bb$
 5. $bC \rightarrow bc$
 6. $cC \rightarrow cc$
- derive the string abc [4 marks]
- (b) Give 3 general naming conventions in programming [5 marks]
- (c) Carry out short circuit evaluation and tabulate the results with the variables A, B [6 marks]
- 5 (a) Write a C++ language program to find the area of a rectangle [9 marks]
- (b) Distinguish between dynamic and static scope in programming [6 marks]
- 6 (a) Identify 3 components of a generative grammar [6 marks]
- (b) Show 2 examples of overload operators in each of (i) Java (ii) C++ (iii) C# programming [6 marks]
- (c) Define a Java literal and show two (2) types of it with an example of each [3 marks]



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2023_2 EXAMINATIONS**

COURSE CODE: CIT332

COURSE CREDIT: 3

COURSE TITLE: SURVEY OF PROGRAMMING LANGUAGES

TIME ALLOWED: 2½ HOURS

INSTRUCTION: ANSWER QUESTION ONE (1) AND ANY OTHER THREE (3) QUESTIONS

QUESTION

1(a) Given the tasks of resetting, fetching, executing and decoding instructions, draw up a top- approach of doing this by a computer. [4 marks]

(b) What are the 4 major strength of the fifth generation of programming languages [5 marks]

(c) Identify and describe the main features List and Tuples in Python programming language [5 marks]

(d) Identify 3 main differences between the python list and python tuple [3.5 marks]

(e) Illustrate the syntax of Class in Java programming and show an example with a class named MyData having one attribute named “b” and one method named “okay” which returns a null value and print a message ”I am alright” from the console. [7.5 marks]

2 (a) Apply maintenance, usage and speed of processing to tabulate the differences between low level and high level languages [7.5 marks]

(b) Analyse the trade-off between each of the following in a medical diagnosis system (i) Reliability and programmability (ii) maintainability and Availability [7.5 marks]

3(a) Show the syntax of the If-else statement in (i) C++ (ii) C# [9 marks]

(b) Analyse the structure PERL programming language [4 marks]

(c) Why is python a portable programming language ? [2 marks]

4 (a) The table below shows the characteristics of programming languages and the criteria for achieving those characteristics. Check the box with an X to show which criterion meets which characteristics [6 marks]

Characteristics	Readability	Writability	Reliability
Simplicity		X	
Orthogonality			
Syntax Design			
Type checking			
Exception Checking			
Expressivity			

(b) Explain 3 types of errors that can occur during the execution of computer program [6 marks]

(c) With the aid of their syntax differentiate between the dotimes and the dolist iteration in LISP [3 marks]

5 (a) Justify the use of HMTL for the development of a web software [3 marks]

(b) Differentiate between ATOM evaluation and LIST evaluation in LISP programming language [6 marks]

(c) Explain the Java switch-case statement and illustrate its syntax [6 marks]

6 (a) Fill the missing information about a generative grammar in the following table

Grammar Type	Grammar Accepted	Language Accepted	Automation
Type 0	Unrestricted grammar		
Type 1			
Type 2			
Type 3			Finite state automaton

[10 marks]

(b) Explain the term syntactic ambiguity

[5 marks]



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2022-1 EXAMINATION

COURSE CODE: CIT 332

COURSE TITLE: Survey of Programming Languages

CREDIT: 4 Units

TIME ALLOWED: 3Hours

INSTRUCTION: Answer Question ONE (1) and any other FOUR (4) Questions

QUESTION ONE

- a. Define the following terms
i. program ii. Algorithm **(4 marks)**
- b. What is stepwise refinement **(2 marks)**
- c. List the aims guiding the design of a good computer program **(3marks)**
- d. List the different stages involved in program development **(3marks)**
- e. What are processing programs? Give examples of these in an operating system. **(4 marks)**
- f. What is the difference between macro flowcharts and micro flowchart? **(2 marks)**
- g. How are languages translated into machine code? **(4 marks)**

(22 marks)

QUESTION TWO

- a. What is Machine Language **(2marks)**
- b. State 2 advantages and 2 disadvantages of Machine Language **(4 marks)**
- c. What is Assembly Language **(2marks)**
- d. State 2 advantages and 2 disadvantages of Assembly Language **(4 marks)**

(12 marks)

QUESTION THREE

- a. What does the acronym BASIC stands for? **(2marks)**
- b. Name the two major variable types in BASIC? **(2marks)**
- c. Write the general format of a BASIC statement. **(2 marks)**
- d. Write a BASIC code that accepts two numbers A, B as input, computes the average of the input as C, and output the result (average). **(6marks)**

(12 marks)

QUESTION FOUR

- a. What does the acronym FORTRAN stand for? **(2marks)**
- b. List examples of FORTRAN 77 compilers? **(2 marks)**

- c. Identify the valid FORTRAN variables and the invalid (with reason) from the following
C\$, 4AB, k4, JOS **(2marks)**
- d. Write a FORTRAN code that accepts two Integers, computes the sum of the two numbers and output the result. **(6 marks)**

(12 marks)

QUESTION FIVE

- a. What are the two basic parts a PASCAL code contains? **(2marks)**
- b. State how to express the exponentiation operation and tangent function in Pascal since there are no direct ways of expressing them **(2marks)**
- c. Write a PASCAL code that accepts the radius of circle, computes the circumference of the circle and outputs the result (circumference). **(8marks)**

(12 marks)

QUESTION SIX

- a. What does the acronym HTML stand for? **(2marks)**
- b. Write the general format of a HTML page. **(4 marks)**
- c. What are the functions of the HTML tags in (b)? **(4 marks)**
- d. Give four (4) examples of web browsers. **(2marks)**

(12 marks)

QUESTION SEVEN

- a. Define the following Visual Basic terms: i. Project ii. Form iii. Object **(6 marks)**
- b. Distinguish between SDI and MDI and give an example each of the applications that allow Form windows in their mode. **(4 marks)**
- c. State at least 4 controls you can place on your forms. **(2 marks)**

(12 marks)



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2023_1 POP EXAMINATION

COURSE CODE: CIT 332

COURSE TITLE: Survey of Programming Languages

CREDIT: 4 Units

TIME ALLOWED: 3Hours

INSTRUCTION: Answer Question ONE (1) and any other FOUR (4) Questions

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COURSE CODE: CIT 332

COURSE TITLE: Survey of Programming Languages

CREDIT: 3 Units

TIME ALLOWED: 3 Hours

INSTRUCTION: Answer Question ONE (1) and any other THREE (3) Questions

QUESTION ONE

- a. What is a programming language? *(2 marks)*
- b. List the major component of each computer programming generation *(5 marks)*
- c. List four common primitive data type *(2marks)*
- d. What is the full meaning of the following acronyms with respect to the Von Neumann architecture *(5marks)*

Acronmyn	Meaning
PC	
CIR	
MAR	
MDR	
IR	

- e. Name two languages in each of the following categories: Von Neumann, Functional, object oriented. *(3marks)*
- f. What is the output of the following code? Show the value of i after each iteration *(6marks)*
For ($i=0$, $i<6$, $i++$)
- g. Define Syntactic ambiguity *(2marks)*

QUESTION TWO

- a. Describe the Von Neumann architecture *(5marks)*
- b. Outline the fetch/decode part of the fetch/decode/execute/reset cycle. *(5 marks)*
- c. State five criteria for language evaluation *(5 marks)*

QUESTION THREE

- a. What are the three general methods of implementing a programming language *(6marks)*
- b. State the compilation process*(4marks)*
- c. Itemize the reasons responsible for program languages concept*(5marks)*

QUESTION FOUR

- a. What is a scripting Language *(2marks)*
- b. State the two approaches in software development *(4marks)*
- c. Explain three application domains of a programming language *(9marks)*

QUESTION FIVE

- a. What is a Parser *(2marks)*
- b. State the types of parsers *(4marks)*
- c. Outline three Functions of Semantic Analysis *(3marks)*
- d. State two differences between syntax and semantics *(4marks)*
- e. Mention the two most common parameter passing techniques *(2marks)*

QUESTION SIX

- a. What is a Subroutine *(2marks)*
- b. State two advantages of a subroutine *(2marks)*
- c. Explain the three types of control structures *(6marks)*
- d. In a tabular form, differentiate between a low level and a high-level language. *(5marks)*